

Methodology

This chapter presents the methodology for reconstructing three-dimensional LV geometry from sparse SAX contours and measuring myocardial wall thickness on the resulting surfaces. The approach combines synthetic geometric supervision with clinical MRI-derived data, a phase-conditioned implicit representation, and a comparison of local wall-thickness estimators.

3.1. Overview

Sparse SAX contours constrain the LV only on the acquired image planes, leaving the intervening geometry to be inferred. Because clinical datasets do not provide independent three-dimensional surfaces for direct supervision, the training strategy first learns complete LV geometry from synthetic SSM samples and then adapts this prior to segmentation-derived clinical meshes. Both stages use the same contour representation, allowing the model to receive a consistent form of sparse observation throughout training.

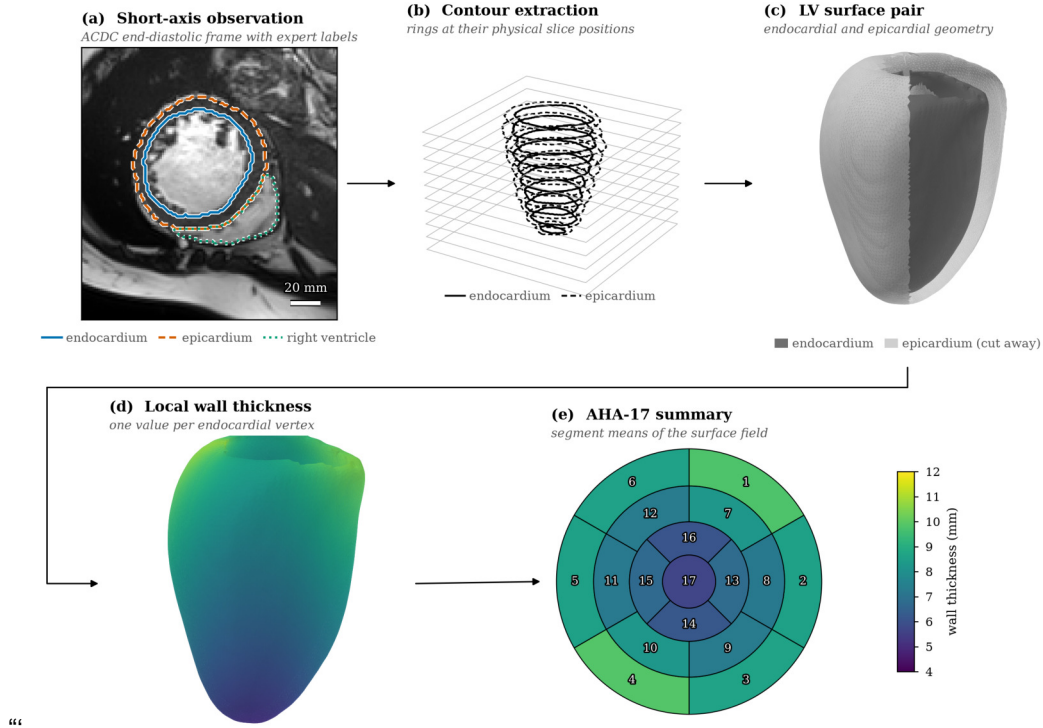


Figure 3.1. Methodology from sparse SAX observations to reconstructed LV surfaces and wall-thickness outputs. ACDC data are shown in panels (a)–(b); panels (c)–(e) use illustrative SSM-derived geometry rather than reported results.

Reconstructing the endocardium and epicardium as unrelated surfaces could produce intersections or local wall collapse. The model therefore uses a phase-conditioned implicit decoder that reconstructs the two boundaries with an explicit positive separation. After surface extraction, several local wall-thickness estimators are compared because the resulting measurement depends not only on the reconstructed geometry, but also on how correspondence across the myocardial wall is defined. Regional AHA-17 aggregation is performed only after the local measurement method has been selected.

Figure 3.1 summarises the complete workflow. The synthetic epicardial offset shown in panels (c)–(e) is used only to illustrate the geometric reconstruction and measurement stages and does not represent a clinical wall-thickness reference.

The chapter follows this dependency. Section 3.2 establishes the synthetic and clinical supervision, Section 3.3 defines the contour encoder and coupled decoder, and Section 3.4 describes model optimisation and surface extraction. Section 3.5 then evaluates the local thickness estimators and regional aggregation. Since the clinical meshes are derived from segmentations rather than acquired independently, the evaluation assesses geometric completion and computational measurement consistency rather than anatomical ground truth.

3.2. Data Preparation

A fundamental challenge in training the model is the absence of a paired dataset that provides both sparse SAX contour observations *and* independently acquired three-dimensional LV meshes. Clinical cardiac MRI datasets supply segmentation masks and image volumes, but they do not include the watertight endocardial and epicardial surfaces that an SDF decoder requires as supervision. This absence of independent three-dimensional targets is the starting point for the entire data-preparation strategy.

The training data were therefore constructed in two stages. First, 1300 synthetic ED geometries were generated from the UK Digital Heart Project LV SSM for pre-training. These samples provide controlled variation consistent with the shape distribution represented by the SSM. Second, ED and ES segmentations from ACDC and M&Ms-2 were converted into watertight meshes for fine-tuning, introducing patient-specific anatomy, acquisition variability, and cardiac-phase variation. The clinical meshes are segmentation-derived surrogate targets rather than independent anatomical ground truth.

Both stages use a common cache schema, which keeps the model interface and loss formulation unchanged as training moves from synthetic to clinical data. The schema also separates the sparse contours supplied to the encoder from the mesh-derived supervision used only during training.

3.2.1. Absence of independent 3D targets and motivation for SSM

The synthetic and clinical streams address different parts of the supervision gap. Sampling the SSM supplies complete, watertight surfaces over a controlled range of LV geometry, but it cannot reproduce scanner-dependent acquisition, clinical label variation, or systolic anatomy. Conversely, the ACDC (Bernard et al., 2018) and M&Ms-2 (Martín-Isla et al., 2023) labels contain patient-specific ED and ES observations, but their slice-wise segmentations must be converted into closed surfaces before they can supervise the implicit decoder. Synthetic pre-training and clinical fine-tuning are used in that order so

that the model first learns a regularised geometric prior and then encounters the conditions present in the evaluation data.

3.2.2. Statistical Shape Model of the left ventricle

The synthetic pre-training data were generated from the UK Digital Heart Project LV SSM reported by Bai et al. (2015), built from more than 1,000 high-resolution cardiac MR images of healthy subjects. The model provides a mean endocardial surface of 22 043 vertices together with 100 principal modes of variation stored as a $66\,129 \times 100$ matrix of mode vectors and a corresponding vector of 100 eigenvalues λ_i . Following the standard SSM formulation (Heimann & Meinzer, 2009), a new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \quad (3.1)$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \dots, b_{100}]^\top$ is the sampled coefficient vector. The standard deviation $\sigma_i = \sqrt{\lambda_i}$ sets the scale of variation along mode i .

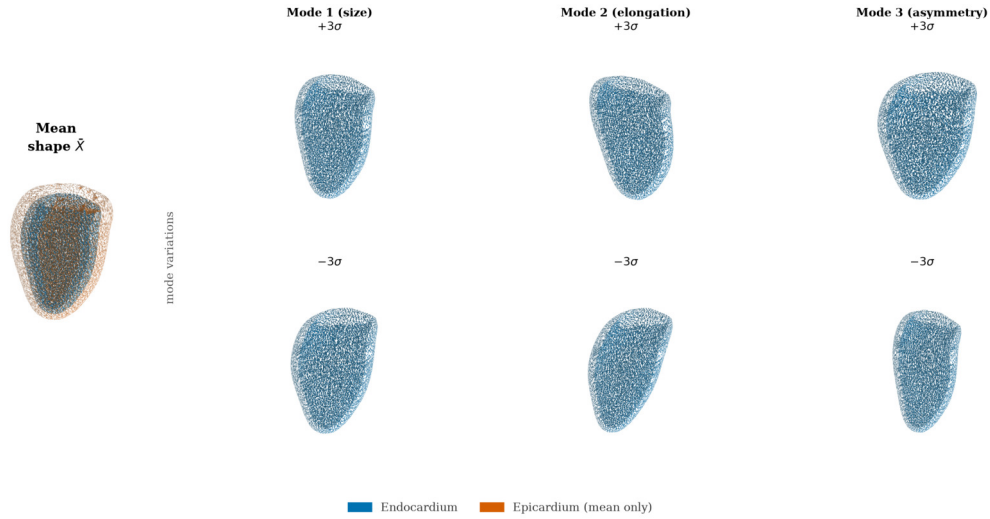


Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

The first few modes capture the largest sources of shape variation in the population; mode 1 primarily controls overall LV size, mode 2 encodes elongation along the long axis, and higher modes represent progressively more localised deformations, as shown in Figure 3.2.

3.2.3. Synthetic end-diastolic dataset (SSM pre-training)

The synthetic ED dataset was built by sampling 1300 meshes from the SSM and converting each mesh into a training-compatible sample. The generation pipeline proceeds in five steps, illustrated in Figure 3.3.

Step 1: Latent coefficient sampling. Each candidate coefficient vector b was drawn by sampling each component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and then applying two acceptance criteria:

$$\left| \frac{b_i}{\sigma_i} \right| \leq 3, \quad \sum_i \left(\frac{b_i}{\sigma_i} \right)^2 \leq \chi_{0.99, d}^2, \quad (3.2)$$

where σ_i is the standard deviation of mode i , d is the number of active modes, and $\chi_{0.99,d}^2$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The per-mode clipping prevents individual modes from dominating the shape, and the Mahalanobis bound prevents the combined deformation from falling outside the plausible population distribution. Accepted samples are scattered within a bounded ellipsoid in latent space, as shown in panel (c) of Figure 3.3.

Step 2: Mesh generation. Each accepted coefficient vector b is substituted into Equation (3.1) to produce a deformed endocardial vertex array, which is then triangulated using the face connectivity of the mean-shape mesh. The resulting triangular surface is watertight and oriented by construction.

Step 3: Quality filtering. Before a generated mesh is accepted into the training set, it is evaluated against four proxy metrics: enclosed volume (mL), surface area (cm²), sphericity index, and plausibility of the apical and basal normal directions. Samples that fall outside calibrated bounds for any of these metrics are discarded. Periodic exact audits using VTK mass-properties computation were also performed every 40 samples to verify the proxy metric calibration.

Step 4: Synthetic epicardium. The SSM provides only the endocardial surface. The epicardial surface was synthesised by offsetting each endocardial vertex p_i along its outward normal n_i by a spatially varying distance d_i ,

$$q_i = p_i + d_i n_i, \quad (3.3)$$

where d_i was set to approximately 10 mm near the base and 5 mm near the apex, with additive Gaussian noise of standard deviation 0.5 mm to avoid perfectly uniform geometry. This synthetic epicardium is a geometric pre-training prior only. It is not intended to reproduce clinical wall thickness, and it is never used as a reference for any wall-thickness result. Its purpose is to give the decoder a consistent outer surface during pre-training, so that the model learns the general shape relation between an inner and an outer LV surface before it sees real data.

Step 5: SAX contour extraction and volumetric supervision points. Each pair of endocardial and epicardial surfaces was sliced at 10 evenly spaced axial levels to produce sparse SAX contour rings, matching the acquisition pattern of real cine MRI. For each case, 2048 query points were sampled near the surfaces and in the surrounding volume, and signed-distance targets or inside/outside labels were computed. These query points form the dense volumetric supervision used by the SDF loss terms during training.

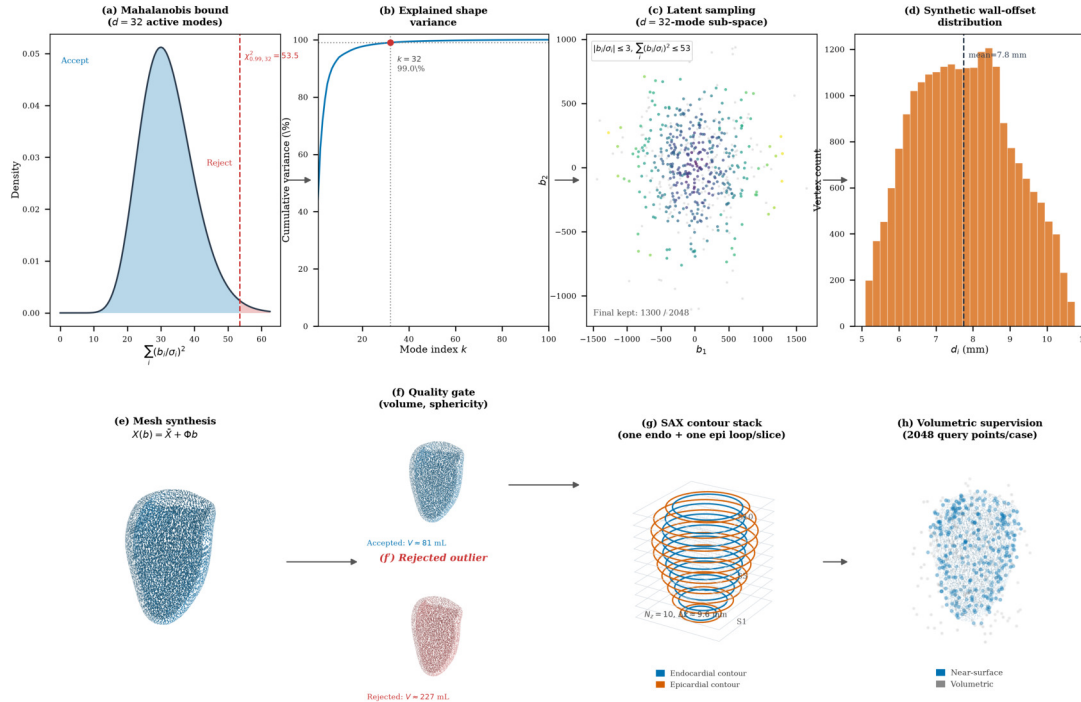


Figure 3.3. Synthetic ED dataset generation from SSM sampling to geometric supervision. Top row: statistical constraints; bottom row: geometric construction and supervision targets.

Figure 3.3 summarises this process in two rows. The top row provides the statistical view of sampling: panel (a) shows the Mahalanobis-statistic density and the $\chi^2_{0.99,32}$ acceptance threshold, panel (b) shows that 32 active modes capture 99% cumulative shape variance, panel (c) shows accepted latent samples in the first two dimensions, and panel (d) shows the synthetic epicardial-offset distribution from Equation (3.3). The bottom row provides the geometric view: panel (e) shows mesh synthesis via Equation (3.1), panel (f) contrasts an accepted shape with a rejected outlier, panel (g) shows the 10 SAX contour levels, and panel (h) shows the 2048 near-surface and volumetric query points used for decoder supervision.

These fixed SSM-sampling and supervision settings were kept constant across all synthetic ED cases to ensure reproducible pre-training before fine-tuning on real ED and ES data.

3.2.4. Real ED and ES labels: from SAX segmentation to 3D supervision

SSM-derived meshes alone cannot capture the full diversity of clinical acquisition conditions: scanner differences, segmentation quality, image resolution, and the distinct geometry of the end-systolic phase. A second data stream was therefore created by reconstructing watertight three-dimensional LV meshes directly from the segmentation masks provided in the ACDC (Bernard et al., 2018) and M&Ms-2 datasets, taking both the ED and ES frames so that the model is exposed to both cardiac phases.

These real ED and ES caches are used in the second optimisation stage, after the initial SSM-based training. The reason for this fine-tuning stage is that the synthetic corpus teaches a strong global LV-shape prior, but it does not fully reproduce clinical variability in contour quality, inter-dataset label style, and phase-specific contraction patterns. Fine-tuning on real ED and ES therefore reduces domain shift

between synthetic supervision and the target clinical setting while preserving the geometric regularity learned from the SSM.

Operationally, the fine-tuning stage keeps the same model architecture and the same multi-term loss from Section 3.4, but replaces the synthetic-only stream with real-derived cache samples from ED and ES frames. Because the real caches follow the same schema (contours, tissue labels, phase label, mesh supervision, and query points), the training loop can continue from the pre-trained checkpoint without changing model interfaces. The phase label then provides the conditioning signal that allows one shared model to adapt from diastolic to systolic geometry during continued training.

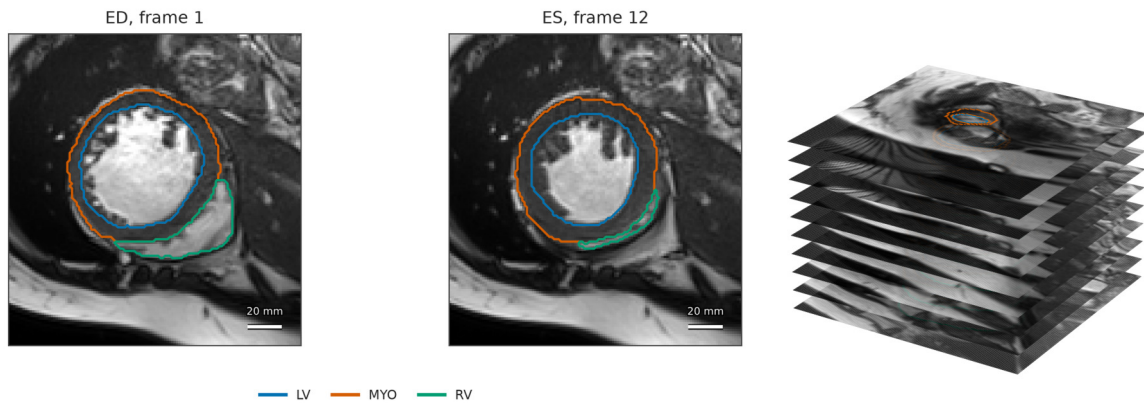


Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

Before describing how these meshes are built, it is useful to look at the raw clinical input, because it makes clear why a reconstruction step is needed at all. Figure 3.4 shows representative SAX slices from the ACDC dataset (Bernard et al., 2018), with the LV cavity, myocardium, and right-ventricular contours overlaid. Each frame is only a two-dimensional cross-section of the heart, and the segmentation is defined one slice at a time. Panel (b) of Figure 3.4 stacks the segmented ED slices at their physical through-plane positions and exposes the central difficulty of the problem: a clinical acquisition provides only a small number of parallel SAX slices separated by several millimetres, so the observed geometry is a set of sparse contour rings rather than a continuous surface.

The reconstruction pipeline closes this gap by converting each sparse label stack into a derived, watertight three-dimensional mesh. These meshes are *segmentation-derived surrogate targets*, not manually annotated or independently acquired 3D reference geometry: their geometry is determined by the observed labels and by the explicitly stated resampling, smoothing, and capping operations. This distinction is important when interpreting the later reconstruction-consistency results. An earlier ES pipeline attempted to fit the SSM to systolic contours; this was abandoned because forcing a diastolic statistical prior onto systolic geometry produced anatomically unrealistic surfaces with collapsed walls. The direct reconstruction approach instead retains the patient-specific geometry present in the ED or ES labels. The two parallel endocardial and epicardial paths are shown in Figure 3.5, and the four stages of the conversion are described below. Table A.2 of Chapter A lists the individual operations in order.

From label volume to clean binary masks. Each segmentation volume is read from NIfTI format and reoriented to the RAS+ convention using its affine, which places the voxel axes in a consistent patient frame while preserving the physical spacing. This step does not define the cardiac long axis. The acquired SAX stack supplies the through-plane direction, and base and apex are identified later from the cross-sectional areas at the two ends of the stack. The ACDC and M&Ms-2 datasets use compatible but not identical label encodings for the LV cavity, the myocardium, and the right ventricle, so each source is first mapped to a common convention. Two binary masks are then formed. The endocardial mask is the LV cavity label, and the epicardial mask is the union of the cavity and the myocardium:

$$M_{\text{endo}} = \mathbb{K}[L = L_{\text{LV}}], \quad M_{\text{epi}} = \mathbb{K}[L \in \{L_{\text{LV}}, L_{\text{MYO}}\}], \quad (3.4)$$

where L is the label image, L_{LV} is the LV cavity label, and L_{MYO} is the myocardium label. The volume is then resampled to 1.5 mm isotropic spacing with nearest-neighbour interpolation, which avoids blending one label into another. Within each slice, the largest connected component of each label is kept and any holes are filled, and the largest three-dimensional component is kept afterwards. This removes the small disconnected fragments that often appear near the first and last slices of the stack. The endocardial mask is finally merged into the epicardial one so that $M_{\text{endo}} \subseteq M_{\text{epi}}$ holds everywhere.

From masks to a smooth closed surface. Applying marching cubes directly to voxel labels produces a visible staircase, so the binary mask is first smoothed with a Gaussian filter at a scale of 1.5 mm and the surface is taken at its 0.5 isosurface (Lorensen & Cline, 1987). Panels (d) and (d') of Figure 3.5 show the unsmoothed voxel surface for the endocardial and epicardial masks, which makes this artefact explicit. Taubin λ/μ smoothing (Taubin, 1995) (25 iterations, $\lambda = 0.53$, $\mu = -0.55$) then removes most of the remaining staircase while approximately preserving the enclosed volume. Because the labels stop at the valve plane, marching cubes closes the top of the ventricle with a rounded dome that does not exist in the anatomy. The upper 6% of the apex-to-base extent is therefore cut away, the opening is closed with a flat cap perpendicular to the long axis, and three iterations of cotangent-Laplacian fairing smooth the join. Quadric-error decimation (Garland & Heckbert, 1997) finally reduces every mesh to at most 4000 vertices, so that all cases reach the cache at a comparable size.

Plausibility checks. Marching cubes reproduces whatever surface the labels imply, even when that surface is not a plausible ventricle. A label stack can be incomplete, noisy, or come from a long-axis rather than a short-axis acquisition, and the resulting mesh is then valid geometry but poor anatomy. Seven checks are therefore applied before a mesh enters the cache, and a case that fails any of them is discarded, because training on implausible geometry would weaken the shape prior the model is meant to learn.

Three of the checks ask whether the stack really covers the ventricle. The cross-sectional area of the endocardium must decrease from the mid-cavity towards the apex, because the ventricle narrows in that direction; when the apical end is the wider one, base and apex have most likely been swapped, or the stack stops before the true apex. The most basal slice must have an area consistent with the valve-plane region, since a very small basal contour means the acquisition ended too early and the basal cap

would then close the mesh at an arbitrary height. The ratio between the apex-to-base length and the largest short-axis diameter must stay inside a fixed range, which removes globally distorted shapes.

The remaining four checks ask whether the segmentation is internally consistent. The slice centroids must not jump sideways from one slice to the next, because that pattern points to motion or an orientation error and produces twisted surfaces. Within each slice, the endocardial and epicardial rings must stay smooth and roughly convex, since jagged or self-intersecting rings usually come from local segmentation failures that smoothing cannot repair. At every level the endocardium must lie inside the epicardium with a positive gap between them, because the opposite ordering is an impossible target for a decoder that models the wall as an outward offset. The average gap between the two rings must also stay within plausible limits, which removes cases where labels have merged or leaked into neighbouring tissue.

Contour and query-point extraction. The encoder receives contours extracted directly from the acquired clinical slices, not contours re-sliced from the derived mesh. For each available SAX slice, the largest endocardial and epicardial label contours are retained and resampled to 60 evenly spaced points. Their physical slice positions are kept, so the encoder receives the original sparse observation in its measured through-plane arrangement. This preserves the acquisition geometry and prevents the derived mesh from leaking into the encoder input.

The derived endocardial and epicardial meshes are centred and normalised with the same translation and scale before they are used as three-dimensional supervision. For each mesh pair, 2048 query points are drawn near the two surfaces and across a padded bounding box. Near-surface points teach the decoder where the surfaces lie; the volume points teach whether a location is inside or outside each surface. Every query point receives occupancy labels for both surfaces. The contours are therefore the sparse input from the clinical slices, whereas the query points and mesh samples are training-only targets derived from the mesh.

Figure 3.5 provides a visual summary of the conversion from clinical labels to mesh-ready supervision.

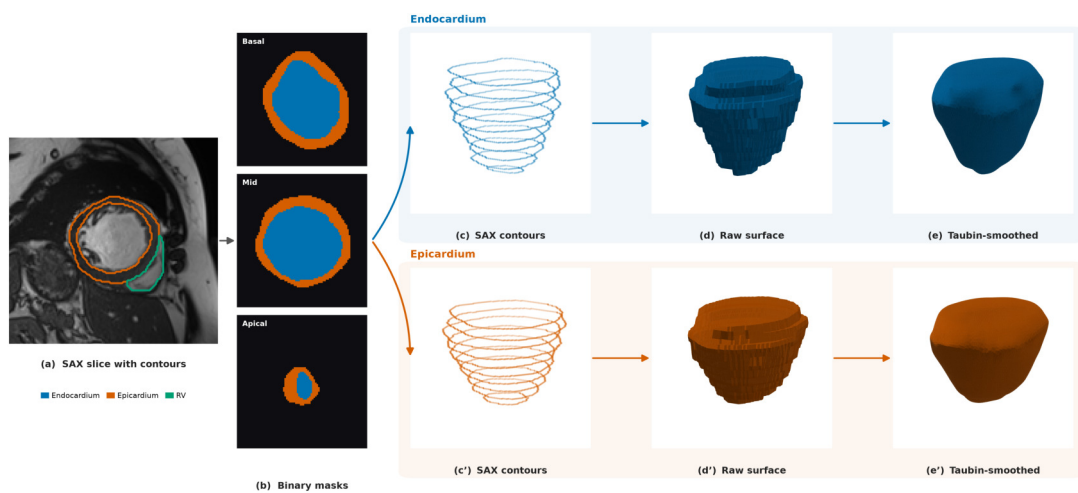


Figure 3.5. Real ED clinical-label-to-mesh pipeline (ACDC): from SAX labels and binary masks to endocardial and epicardial surface branches used for derived supervision.

Panel (a) shows the acquired SAX image with LV cavity, myocardium, and RV labels, and panel (b) shows the corresponding binary endocardial and epicardial masks after label harmonisation. The pipeline then branches into separate endocardial and epicardial paths: panels (c)–(e) and panels (c′)–(e′), respectively.

In both branches, the voxel surface view makes discretisation artefacts explicit, and Taubin smoothing illustrates their reduction before later post-processing. The figure is intentionally a process overview; the full sequence also includes basal capping, cap fairing, decimation, and the plausibility checks. Contours for the encoder remain those extracted from the original clinical slices.

3.2.5. Shared cache representation and contour augmentation

All cases are stored in a common cache representation so that the model receives the same interface regardless of whether a sample originates from the SSM or from a clinical segmentation.

A cache item combines the sparse contour observation with mesh-derived supervision and the metadata needed to keep both representations in a shared coordinate frame. The ED and ES caches contain 10 SAX levels, 60 points per available contour ring, and 2048 query points per case. This design deliberately separates what the model observes from what it is allowed to learn from: the encoder receives only contour coordinates, tissue identity, and cardiac phase, whereas the decoder is supervised by surface and volumetric samples derived from the corresponding mesh. Thus, at inference time, the same model can reconstruct geometry from sparse contour observations alone.

Table 3.1. Fields of one shared cache item and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical slice labels	Sparse SAX observation for the encoder and contour-anchor loss
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and normal-alignment supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense SDF, occupancy, sign, and exterior supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Table 3.1 makes this separation explicit. The first three fields describe the clinical-style observation supplied to the encoder; the remaining fields are available only during training, evaluation, or visualisation. Keeping these roles distinct prevents mesh-derived information from leaking into the encoder input while allowing dense supervision of the implicit field.

This common contract enables two-stage optimisation without dataset-specific changes to the architecture or loss. The model is first trained on synthetic ED samples, which establish a broad geometric

prior, and is then fine-tuned on real ED and ES samples, which introduce patient-specific anatomy and phase-dependent shape variation. The loader changes only the source stream between stages; the fields consumed by the training loop remain unchanged.

Contour augmentation is applied to both synthetic ED and real ED/ES training observations, after the train/validation/test split. The target mesh, surface samples, query points, occupancy labels, tissue labels, and phase label remain fixed. This is therefore an observation-robustness intervention rather than the creation of a geometrically transformed anatomical target. Such label-preserving perturbations are commonly used to improve robustness when labelled data are limited (Chlap et al., 2021; Shorten & Khoshgoftaar, 2019).

For each augmented sample, the loader draws small independent in-plane translations per slice, point-wise in-plane jitter, global scale jitter, and contour-point dropout. Slice dropout and long-axis rotation are applied probabilistically while retaining at least three slice levels. The endocardial and epicardial contours within a slice receive the same translation, preserving their local relationship. As illustrated in Figure 3.6, these perturbations ask the encoder to recover one fixed mesh-derived target from plausible variations of its sparse SAX observation.

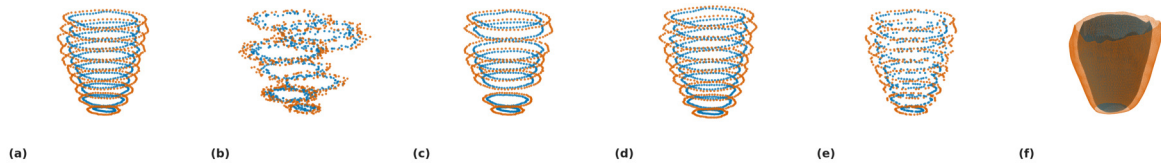


Figure 3.6. Examples of contour-input augmentation. Endocardial and epicardial points are shown in blue and orange, respectively.

In Figure 3.6, panel (a) shows the unaltered contour stack, while panels (b), (d), and (e) show slice translation, slice rotation, and contour-point dropout, respectively. Panel (f) shows the corresponding 3D mesh label. Thus, the sparse SAX observation changes across the augmented examples, whereas the derived mesh and all decoder-supervision targets remain fixed; the figure illustrates robustness of the encoder input rather than transformation of the anatomical ground-truth geometry.

3.2.6. Dataset composition and splits

This subsection reports how many samples each data stream finally contributes and how the real streams are divided for training and evaluation. The real data come from two public collections: the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2). Together they provide 460 patients (100 from ACDC and 360 from M&Ms-2). For each patient we reconstruct one ED mesh and one ES mesh, giving two real streams. Both collections are openly available for research at no cost; each requires a free registration and the acceptance of a data-use agreement. The original M&Ms challenge paper (Campello et al., 2021), discussed in Chapter 2, describes a separate 375-study dataset and is not used as a citation for the M&Ms-2 release. The synthetic stream is generated from the UK Digital Heart Project SSM: of its 100 modes, the

first 32 already capture 99% of the shape variance, so we sample within that sub-space. Drawing shape coefficients under the Mahalanobis bound of Equation (3.2) and rejecting implausible shapes yields 1300 accepted meshes out of 2048 candidates (63%). The shape model is likewise open and free: it is distributed as a public repository by the UK Digital Heart Project,¹ needs no access application, and the authors ask only that the atlas papers be cited. None of the three sources used in this work therefore required a licence fee.

Not every real patient produces a usable mesh. A segmentation may be a long-axis view, cover too few slices, or fail an anatomical plausibility check, such as an endocardium that is not fully enclosed by the epicardium. After these quality gates, 444 of the 460 patients yield a valid ED mesh and 423 yield a valid ES mesh. The surviving meshes are then divided for training and evaluation. To prevent information from leaking between the partitions, the real streams are split by patient rather than by mesh, so that all meshes from a given patient stay within a single partition. Both streams use the same 70%/15%/15% allocation for training, validation, and test, rounded to whole patients, and the synthetic ED meshes are used only to enrich the pre-training stage and are never placed in the validation or test sets.

Table 3.2. Final sample counts and partitioning of the three data streams. Real streams report patients that pass the quality gates; the synthetic stream reports meshes accepted out of the 2048 sampled shape vectors. The reported clinical evaluation contains no synthetic cases.

Stream	Source	Phase	Count
Real data	ACDC and M&Ms-2	ED	444 patients
Real data	ACDC and M&Ms-2	ES	423 patients
Synthetic data	UK Digital Heart SSM	ED	1300 meshes

Table 3.2 records the two real source collections and the synthetic pre-training stream. The role of each stream in the two-stage training procedure is summarised in Chapter A.

3.3. Model Architecture

The model is built from two complementary parts. The encoder reads the sparse contour points and summarises the case into a compact latent representation, while a local contour-volume branch keeps the surrounding spatial structure in a coarse 3D feature field. The decoder then combines the global latent code, the local feature at each query point, and the point coordinates to predict the LV endocardial signed-distance field. Because the decoder can be evaluated anywhere in space, the reconstructed surface is not tied to a fixed grid or a fixed number of vertices, and a mesh is only generated once the field has been sampled densely. The output is then a continuous signed-distance representation of the chamber and its surrounding wall.

This design matches the structure of the problem. The contour input is a small, unordered set of points, so the encoder uses a point-cloud formulation that is invariant to the order of the points. The output is a smooth, continuous surface, so the decoder is an implicit function that can be evaluated at any

¹

resolution. The epicardial field is defined through a positive offset from the endocardial field, preventing collapse of the two predicted fields. Anatomical wall thickness is subsequently measured between the extracted surfaces.

Figure 3.7 links the global shape code, local feature volume, and coupled decoder. The offset shown in the figure is a separation between implicit fields, not a measured transmural distance. The encoder is formalised in Section 3.3.1 and the decoder in Section 3.3.2.

3.3.1. Contour encoder

The encoder receives, for each contour point, its 3D coordinates together with a label indicating whether the point is endocardial or epicardial and a label indicating the cardiac phase. A small shared network processes every point independently and turns it into a feature vector. The features of the endocardial points and the features of the epicardial points are then combined separately by taking, for each feature, its maximum over the points. This makes the latent code invariant to the order of the contour points, which is important because contour points have no natural ordering across patients or slices. The two resulting descriptors are joined and projected to a single 256-dimensional latent vector z . At the same time, the same point features are scattered into a local 3D feature volume that keeps spatial neighbourhood information around the LV wall.

More formally, let $P = \{p_i\}_{i=1}^N$ be the contour points of one case, where each p_i carries coordinates, a tissue label, and a phase label. The shared point network computes

$$h_i = \phi_{\text{enc}}(p_i), \quad (3.5)$$

where ϕ_{enc} is the shared network. The endocardial and epicardial descriptors are then obtained by max-pooling within each tissue,

$$z_{\text{endo}} = \max_{i \in \mathcal{E}} h_i, \quad z_{\text{epi}} = \max_{i \in \mathcal{P}} h_i, \quad (3.6)$$

where $\mathcal{E}$ and $\mathcal{P}$ are the endocardial and epicardial point sets. The global latent code is then

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \quad (3.7)$$

while the same scattered features are aggregated into a local 3D volume V and sampled at each query point by trilinear interpolation,

$$v(x) = \text{Interp}(V, x). \quad (3.8)$$

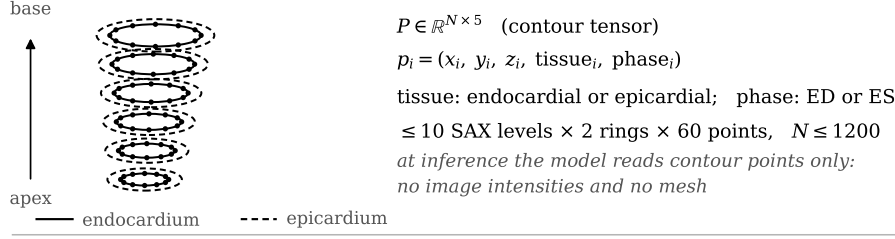
This local field adds spatial detail to the decoder while the global latent vector keeps the overall chamber shape compact and stable.

Separate pooling preserves the distinct roles of the cavity and outer wall, while the local volume supplies spatial context near each query point.

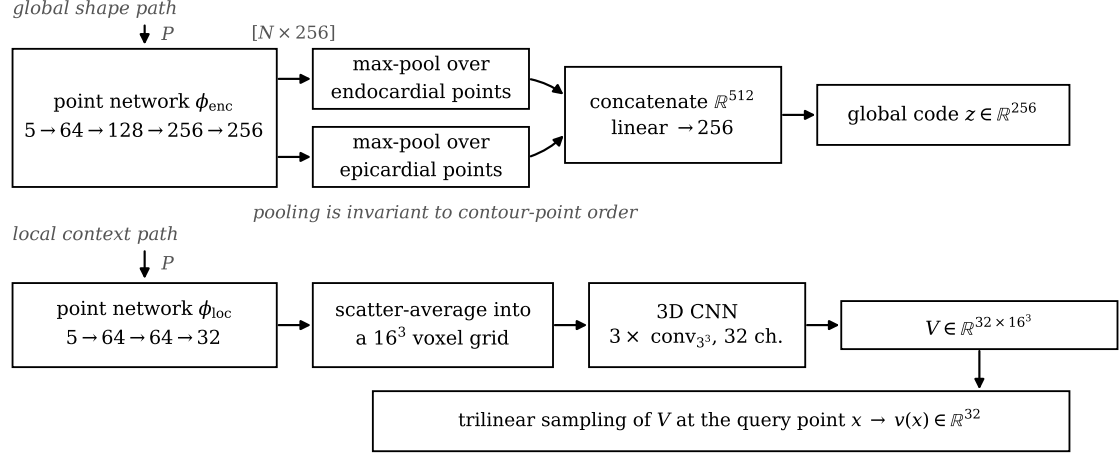
3.3.2. Implicit decoder and positive-wall coupling

Before a query point is given to the decoder, its coordinates are expanded with a Fourier positional encoding. This simply adds a few sine and cosine copies of the coordinates at different frequencies, which helps the network represent fine geometric detail instead of only smooth, blurry shapes (Tancik

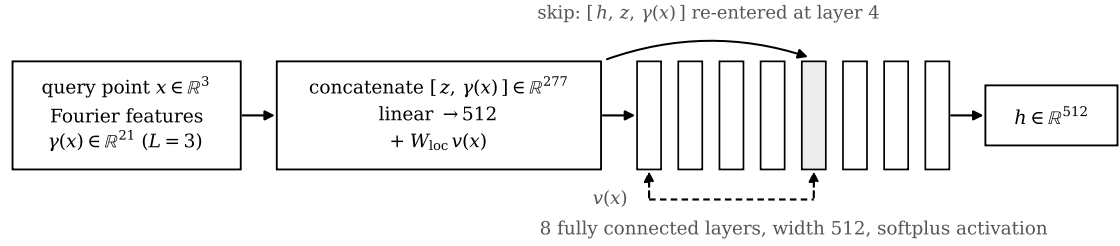
(a) Input: sparse short-axis contour observation



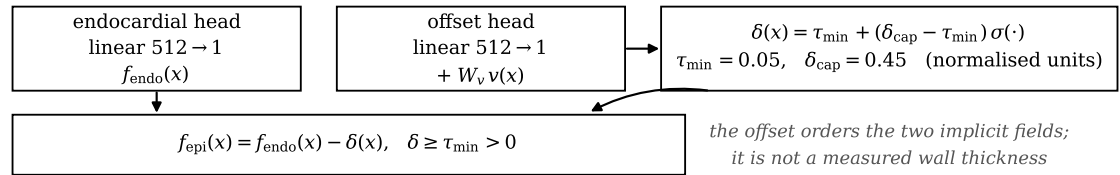
(b) Contour encoder



(c) Implicit decoder



(d) Output heads and field coupling



(e) Surface extraction

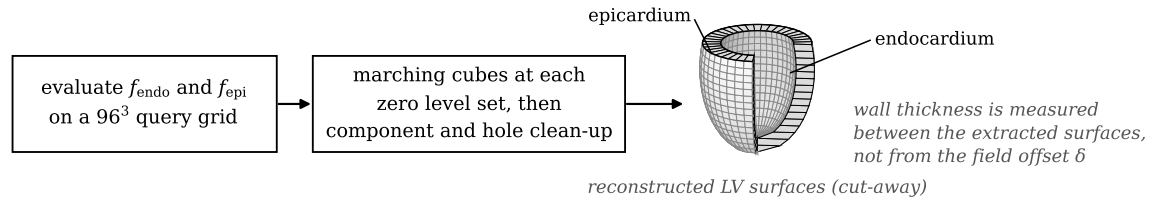


Figure 3.7. Phase-conditioned model architecture. Tissue-specific global and local contour features condition an implicit decoder, which predicts the endocardial field and a positive wall offset before mesh extraction.

et al., 2020). Three frequency bands are used. For a query point $x \in \mathbb{R}^3$ the encoding is

$$\gamma(x) = [x, \sin(2^0 \pi x), \cos(2^0 \pi x), \dots, \sin(2^{L-1} \pi x), \cos(2^{L-1} \pi x)], \quad (3.9)$$

with $L = 3$. The decoder itself is an eight-layer fully connected network of width 512 with a skip connection at the middle layer. It uses a smooth activation (softplus) so that the reconstructed surface is smooth and free of the sharp creases that a ReLU activation would introduce.

The decoder produces two outputs. The first is the signed distance to the endocardium, f_{endo} . The second is a positive decoder wall-offset field δ , which controls the separation of the predicted epicardial and endocardial fields at each query point. It is a learned difference between field outputs, not a Euclidean transmural distance measured between the extracted surfaces. The decoder also receives the local contour feature $v(x)$ from the spatial volume, which is injected into the model so that it can use local wall context when predicting the offset. The offset combines a bounded sigmoid with a soft-hinge headroom term above the sigmoid's saturation knee,

$$\delta(x, z, v) = \tau_{\min} + (\delta_{\text{cap}} - \tau_{\min})\sigma(r) + \frac{h}{\beta_{\text{hr}}} \log(1 + \exp[\beta_{\text{hr}}(r - r_0)]), \quad r = g_{\delta}(x, z, v), \quad (3.10)$$

where g_{δ} is the raw decoder output, σ is the sigmoid, $h > 0$ is a learned headroom slope, $\beta_{\text{hr}} = 8$, and $r_0 = \log(k/(1 - k))$ with $k = 0.99$. The constants $\tau_{\min} = 0.05$ and $\delta_{\text{cap}} = 0.45$ define the lower bound and the ceiling of the sigmoid component in normalised units. The headroom term is negligible below the saturation knee but allows δ to exceed δ_{cap} . The epicardial field is then

$$f_{\text{epi}}(x, z, v) = f_{\text{endo}}(x, z, v) - \delta(x, z, v). \quad (3.11)$$

Because Equation (3.10) always keeps $\delta \geq \tau_{\min} > 0$, the field offset cannot be zero or negative. This prevents a collapsed wall in the field, but it does not establish wall thickness, anatomical correctness, or the topology of the extracted meshes after discretisation. In particular, δ_{cap} is not an upper bound of the complete offset.

The complete architecture configuration is collected in Table A.3 of Chapter A.

3.4. Training and Inference

This section explains how the model is trained from the shared cache and how a surface mesh, decoder offset map, and subsequent wall-thickness measurements are produced once training is finished. Training proceeds in two stages: the model is first pre-trained on the synthetic SSM cases and then fine-tuned on the real ED and ES cases. In both stages it must learn two things together: a valid distance field for the endocardium, and a sensible relationship between the endocardium and the epicardium.

3.4.1. Training data mix and augmentation

Training follows the two-stage strategy introduced in Section 3.2. In the first stage, the model is pre-trained only on the synthetic ED cases generated from the SSM. This controlled corpus provides the model with a broad geometric prior for LV shape and allows stronger augmentation of the sparse observations.

In the second stage, the pre-trained model is fine-tuned on the real cases reconstructed from ACDC and M&Ms-2, covering both the ED and ES phases. Fine-tuning starts from the pre-trained weights and continues with the same architecture and loss, so the model keeps the geometric regularity learned from the SSM while adapting to real segmentation quality and to the systolic phase. Throughout both stages the phase label is provided as input, so a single model handles ED and ES. Contour augmentation is used for both synthetic ED and real ED/ES observations; their surrogate targets and cached supervision are never changed.

A single phase-conditioned model is used instead of two separate models. The main reason is practical. When the experiments reported here were carried out, the public repository of the UK Digital Heart Project contained only the end-diastolic shape model, so a separate end-systolic model could not have benefited from the synthetic pre-training at all. An end-systolic shape model has since been added to that repository, but only after the training described in this chapter had been completed. Sharing one model lets the end-systolic cases reuse the shape prior learned from the synthetic end-diastolic corpus, and the phase label then tells the encoder which of the two configurations it is looking at. The cost of this choice is that the phase information is coarse. The model receives a binary label, not a continuous position in the cardiac cycle, so it can only represent two fixed states rather than the gradual change between them. Since the synthetic corpus is end-diastolic, the prior is also biased towards diastolic shape. Both effects make the end-systolic reconstruction the harder of the two cases, and this is visible in the results of Chapter 4.

The two stages are complementary. The synthetic ED cases give a large set of anatomically valid shapes but do not reproduce every artefact seen in real segmentations, which is why they are used for pre-training rather than as the final training data. Their role is to provide a geometric prior: they regularise the shape space, expose the model to a wide range of plausible LV geometries, and do so under controlled conditions. They do not provide realistic patient-specific wall thickness, because the epicardium in those samples is constructed by an offset rule. The real ED and ES cases then supply patient-specific anatomy from several datasets and let the model learn how the shape changes between the two phases. To avoid information leaking between the training, validation, and test sets, the real cases are split by patient before augmentation. Because augmentation perturbs only the observed contours, each real surrogate target remains tied to the anatomy of its own patient.

In practice, this means the encoder is trained to be robust to imperfect, noisy inputs, such as small shifts, jitter, missing slices, or dropped points, while the decoder is always supervised against a fixed, consistent 3D geometry.

3.4.2. Loss function

The training objective contains five terms. Let $\mathcal{E}$ and $\mathcal{P}$ denote sampled endocardial and epicardial surface points, $\mathcal{Q}$ the cached SDF query points, $\mathcal{R}$ the union of near-surface and free-space points used for gradient regularisation, and $\mathcal{F}$ the free-space subset. At each $q \in \mathcal{Q}$, $y_{\text{endo}}(q)$ and $y_{\text{epi}}(q)$ are the cached signed-distance targets. At each $x \in \mathcal{E}$, the wall target is $t(x) = -y_{\text{epi}}(x)$, the unsigned epicardial

distance evaluated on the endocardial surface. The implemented loss terms are

$$\begin{aligned}
\mathcal{L}_{\text{surf}} &= \mathbb{E}_{x \in \mathcal{E}}[|f_{\text{endo}}(x)|] + \mathbb{E}_{x \in \mathcal{P}}[|f_{\text{epi}}(x)|], \\
\mathcal{L}_{\text{SDF}} &= \mathbb{E}_{q \in \mathcal{Q}}[|f_{\text{endo}}(q) - y_{\text{endo}}(q)|] + \mathbb{E}_{q \in \mathcal{Q}}[|f_{\text{epi}}(q) - y_{\text{epi}}(q)|], \\
\mathcal{L}_{\text{eik}} &= \mathbb{E}_{x \in \mathcal{R}}[(\|\nabla_x f_{\text{endo}}(x)\|_2 - 1)^2] + \mathbb{E}_{x \in \mathcal{R}}[(\|\nabla_x f_{\text{epi}}(x)\|_2 - 1)^2], \\
\mathcal{L}_{\text{off}} &= \mathbb{E}_{x \in \mathcal{F}}[e^{-\alpha|f_{\text{endo}}(x)|}] + \mathbb{E}_{x \in \mathcal{F}}[e^{-\alpha|f_{\text{epi}}(x)|}], \\
\mathcal{L}_{\text{WT}} &= \mathbb{E}_{x \in \mathcal{E}}[\ell_{\beta_{\text{WT}}}(\delta(x) - t(x))],
\end{aligned} \tag{3.12}$$

where $\alpha = 5$ and ℓ_{β} is the Smooth-L1 penalty

$$\ell_{\beta}(a) = \begin{cases} a^2/(2\beta), & |a| < \beta, \\ |a| - \beta/2, & |a| \geq \beta, \end{cases} \quad \beta_{\text{WT}} = 0.02. \tag{3.13}$$

All distances in these terms are expressed in the normalised coordinate system of each case. The complete objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}. \tag{3.14}$$

The surface and dense SDF terms fit the two fields to their geometric targets. The Eikonal term encourages both field gradients to have unit magnitude (Gropp et al., 2020), while the exponential off-surface term suppresses additional zero crossings. The wall term directly supervises δ at endocardial surface points. The Eikonal term is computed in full precision to keep its gradient calculation numerically stable.

Table 3.3. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathcal{L}_{\text{eik}}$	Encourages $\ \nabla f\ = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains separation of the predicted fields

3.4.3. Optimisation and hardware

The retained training run uses AdamW with a learning rate of 2×10^{-5} and weight decay 5×10^{-4} . Its learning rate follows cosine annealing with warm restarts, with an initial restart period of 100 epochs. The synthetic pre-training stage lasted approximately 200 epochs; the recorded real-data fine-tuning history then continued for 776 epochs, from which the checkpoint at epoch 695 was selected by validation loss. Fine-tuning resumes from the pre-trained weights rather than starting from scratch. Training uses mixed

precision and gradient clipping at 1.0. Each of the two GPUs processes a local batch of 8 cases and gradients are accumulated for two steps, giving an effective batch size of 32. A light weight-clipping strategy was used instead of the more expensive spectral normalisation, since the decoder is evaluated at many points per step and GPU memory is the main constraint.

Table 3.4. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged
Optimiser	AdamW
Learning rate	2×10^{-5}
Scheduler	Cosine annealing with warm restarts ($T_0 = 100$ epochs)
Weight decay	5×10^{-4}
Training duration	Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695)
Batch size	8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32)
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3

3.4.4. Inference and decoder-offset extraction

At inference time, the model turns the input contours into the latent vector z and then evaluates the decoder on a regular 96^3 grid of points. Marching cubes reads the zero-level sets of f_{endo} and f_{epi} from this grid and produces the raw endocardial and epicardial meshes (Lorensen & Cline, 1987). For evaluation, degenerate faces are removed, the largest connected component is retained, holes are filled, and a mesh is remade from a cleaned occupancy mask when needed. This post-processing makes the evaluation geometry usable for surface and thickness measurements; it is not a guarantee provided by the neural architecture, and the reported metrics do not describe an entirely unmodified decoder output. A raw analytic offset is available at every endocardial vertex because the decoder predicts δ there. Converting this uncalibrated offset to millimetres gives

$$o_{\text{raw}}(x) = s \delta(x, z, v(x)), \quad (3.15)$$

where s is the scale that maps normalised coordinates back to millimetres. This raw analytic offset is not measured after the fact by comparing two meshes, and it is not necessarily a Euclidean transmural separation. For visualisation, it is separately calibrated per phase by $o_{\text{cal}} = \alpha o_{\text{raw}}$, where α is the ratio between the mean segmentation-distance-transform reference and the mean raw analytic offset. This

calibration fixes the mean magnitude by construction and is not an independent test of anatomical accuracy. Wall thickness is instead estimated on the reconstructed meshes by the methods in Section 3.5; the method judged most consistent in the comparison provides the final regional measurements.

The reconstruction is also compared with a simple non-neural baseline built from exactly the same SAX contour rings. Each ring is represented by 60 points. The starting point of each successive ring is shifted to minimise its mean in-plane distance from the preceding ring, while preserving the contour direction. Corresponding points on adjacent rings are then joined by triangles, and the first and last rings are closed with planar caps. The resulting loft is rasterised on a 1.0 mm isotropic grid and extracted again with marching cubes so that its mesh density is comparable with the other evaluation meshes. It then passes through the same watertight repair and smoothing procedure used for the model output. The model and the lofting baseline are evaluated on the same patients against the same segmentation-derived geometry and with the same surface, overlap, and volume metrics. This comparison tests learned geometric completion against direct linear interpolation between the observed contours; it does not isolate the contribution of individual model components.

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the model-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes. Myocardial wall thickness is a clinically relevant morphological measurement, but the present evaluation does not assess diagnostic performance for infarction, scarring, or hypertrophy. The evaluation is therefore organised in three steps: first the algorithms that measure thickness on the geometry, then the resulting local three-dimensional thickness field, and finally its aggregation into the clinical AHA-17 regional summary.

3.5.1. Selected thickness methods

Four wall-thickness algorithms are evaluated on the reconstructed geometry. They are chosen to span the main methodological families used to measure thickness: volumetric distance fields, ray intersections, and partial-differential-equation (PDE) formulations. The four are the Laplace field method, the Yezzi–Prince Eulerian PDE method, the signed-distance cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. No method is assumed in advance to be the best one. All four are computed on the same geometry and compared under the same conditions, and the one that proves most consistent is then used for the regional analysis reported in Chapter 4. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

Table 3.5. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept only as a baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

The EDT boundary sum computes, at each myocardial voxel x , the sum of the distances to the endocardial and epicardial surfaces,

$$t^{\text{EDT}}(x) = D_{\text{endo}}(x) + D_{\text{epi}}(x), \quad (3.16)$$

where D_{endo} and D_{epi} are the Euclidean distance transforms of the two boundaries. This is exact for planar parallel walls but is limited by voxel resolution and slightly overestimates thickness in strongly curved regions.

The Laplace field method defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2 \psi = 0, \quad \psi|_{\text{endo}} = 0, \quad \psi|_{\text{epi}} = 1. \quad (3.17)$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla \psi \cdot \nabla u = -1, \quad t = u + v, \quad (3.18)$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. Inspired by the shape diameter function (Shapira et al., 2008), for each endocardial point several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\text{cone}} = \text{median}_{k=1}^K d_{ik}, \quad K = 7. \quad (3.19)$$

In the notebook implementation the cone half-angle is 30° . This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

Figure 3.8 contrasts how the four estimators measure thickness on the same wall cross-section. The EDT method sums the distances from an interior point to both boundaries; the Laplace method follows the transmural streamlines of a harmonic potential whose nested iso-surfaces are shown as dashed

lines; the Yezzi–Prince method integrates two travel times u and v along that same field and adds them; and the cone-ray method casts a fan of rays around the endocardial normal and keeps the median hit distance.

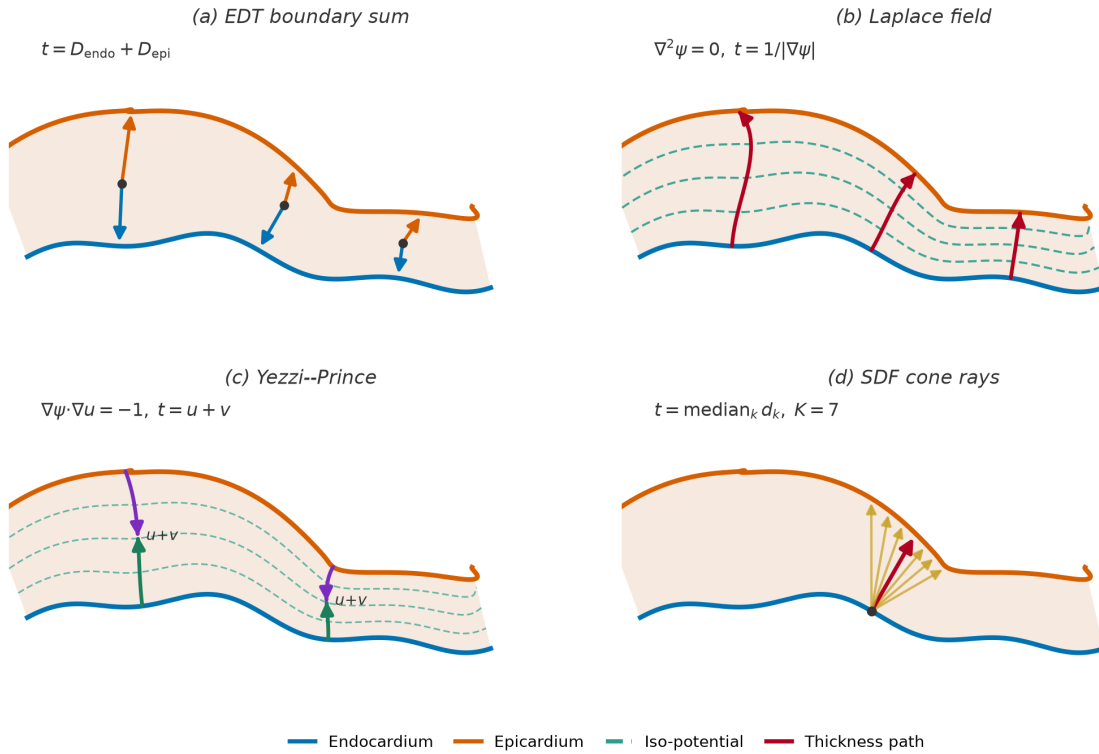


Figure 3.8. Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved transmural streamlines of a harmonic potential, which cross the iso-potential surfaces (dashed lines) at right angles, and estimates thickness from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel times u and v from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

3.5.2. Local three-dimensional wall thickness

Each of the selected methods produces, before any regional summary, a continuous thickness field defined over the whole reconstructed geometry. After the evaluation post-processing described in Section 3.4.4, the endocardial and epicardial surfaces are densely sampled and a thickness value is attached to each endocardial vertex. This produces a local three-dimensional wall-thickness map rather than a handful of per-slice numbers, as shown in Figure 3.9, where the LV surface is coloured by local thickness from the thin apical region to the thicker basal wall.

This local map is the primary output of the wall-thickness stage. Presenting the measurement first as a full surface field, and only afterwards as regional segment averages, makes the spatial distribution of thickness explicit: localised thinning or thickening remains visible on the surface even when it would be averaged away inside a single AHA segment. The colour map uses a fixed thickness range across cases so that maps from different patients and cardiac phases can be compared directly.

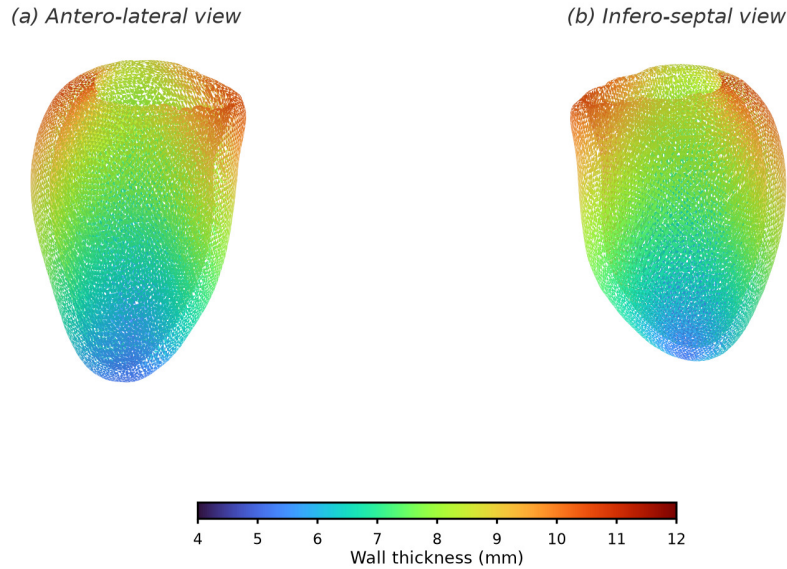


Figure 3.9. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

For the final regional analysis, the surface field from the method judged most consistent in the comparison is aggregated into the AHA 17-segment model, the accepted clinical language for reporting regional myocardial measurements (Cerqueira et al., 2002). Conventionally the wall thickness in each segment is read semi-manually from a few SAX slices, which is observer-dependent, coarsely sampled, and sensitive to the choice of slice and measurement point. Here the same segment values are produced objectively from the selected method's field of Section 3.5.2, by cutting the reconstructed surface into the AHA segments and averaging within each one.

The cut is defined in the LV's own anatomical frame, which first requires identifying the long-axis direction, distinguishing base from apex, and fixing a septal reference direction; the wall-thickness notebook includes an anatomical QA step that visualises this base/apex orientation and septal alignment before the segments are formed. The surface is then partitioned in two stages, as illustrated in Figure 3.10. First, along the long axis, the endocardial vertices are split by their normalised apex–base position into a basal ring, a mid-cavity ring, an apical ring, and a small apical cap. Second, within each ring the vertices are divided angularly around the long-axis centreline: the basal and mid rings are cut into six sectors each and the apical ring into four, while the apical cap forms the single seventeenth segment. Numbering the sectors from the anterior–septal reference and unrolling the rings onto concentric circles yields the familiar bullseye, in which the basal ring is the outer annulus, the mid ring the middle annulus, the apical ring the inner annulus, and the apical cap the central disc.

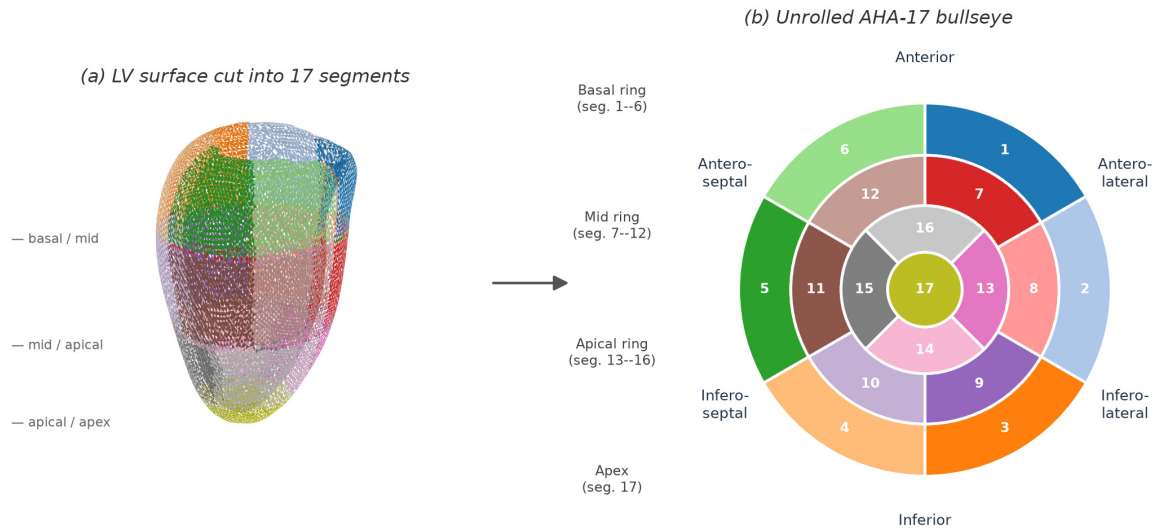


Figure 3.10. Cutting the reconstructed LV surface into the AHA-17 segments. Panel (a) shows the endocardial surface partitioned along the long axis into basal, mid, apical, and apical-cap levels and angularly into sectors, with each of the 17 regions drawn in a distinct colour. Panel (b) unrolls the same colour-coded regions onto the standard AHA-17 bullseye, where the basal ring (segments 1–6) forms the outer annulus, the mid ring (7–12) the middle annulus, the apical ring (13–16) the inner annulus, and the apical cap (17) the central disc; the anatomical wall regions are labelled around the disc.

The AHA aggregation does not replace the per-vertex thickness maps; it provides a clinically familiar regional summary of the same local measurements. For each method, the thickness values of the vertices falling in a segment are combined using robust summaries such as the mean and percentiles, so that local outliers do not dominate the regional interpretation.

This evaluation should be interpreted as an algorithmic comparison and reproducibility study rather than as clinical validation. Unless expert manual wall-thickness measurements and independent 3D targets are added, the thesis can conclude only that the model provides a locally resolved computational framework for reconstruction consistency; it should not claim anatomical truth, diagnostic utility, or replacement of clinical measurement protocols.